



PhishGuard

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HOW IT WORKS

1. User navigates to a website
2. Extension automatically scans:
 - URL structure
 - Page HTML content
3. Suspicious patterns are detected
4. A risk score is generated
5. User receives a warning if the site is unsafe

PHISHGUARD IS A CHROME EXTENSION THAT DETECTS PHISHING WEBSITES IN REAL TIME BY ANALYZING URLS AND WEBPAGE CONTENT. IT PROVIDES USERS WITH IMMEDIATE WARNINGS TO PREVENT CREDENTIAL THEFT AND MALICIOUS INTERACTIONS.

Extension actively scanning a webpage and displaying a phishing risk warning to the user.

CHALLENGES

False Positives

- Legitimate sites (e.g., Google Drive) flagged as suspicious
- → Solution: improved filtering and domain checks

Performance

- Real-time scanning slowed browsing
- → Solution: optimized detection logic and reduced redundancy

Technology

- JavaScript → powers real-time scanning and logic
- Chrome Extension APIs → access browser tabs and URLs
- HTML/CSS → popup UI and warning interface
- Heuristic detection rules → identify phishing patterns

Phishing Detector

Paste a URL here

Test Pasted URL

Safe

Source: Automatic Real-Time Scan

Scanned: 4/16/2026, 4:15:59 PM

Risk Score: 0

URL:

<https://www.instagram.com/>

Reasons:

- No obvious page-structure phishing indicators found

FUTURE WORK

- Machine learning-based detection
- Smarter whitelist system
- Improved UI/UX design
- Expanded database of phishing patterns